

Probability Primer for Supervised, Unsupervised and Reinforcement Learning

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Two Paths

- Rigorous Development / Measure Theory Approach
Probability: Theory and Examples, Richard Durrett
- Intuition / Heuristic Approach
Probability Models, Sheldon Ross
- For rigorous probability proofs, we use the former.
- For practical applications, we use the latter.
- We will focus on the latter.

Preliminaries

- Sample space: S - the set of all possible outcomes of an experiment.
- Event: E - a subset of the sample space.
- Probability of Event E : $P(E)$

Conditions for $P(E)$

- 1) $0 \leq P(E) \leq 1$
- 2) $P(S) = 1$
- 3) For $E_1, E_2, E_3 \dots$ where $E_i \cap E_j = \emptyset, \quad i \neq j$

$$P\left(\bigcup_{n=1}^{\infty} E_n\right) = \sum_{n=1}^{\infty} P(E_n)$$

Properties

$$P(E \cup F) = P(E) + P(F) - P(E \cap F), \quad P(EF) = P(E \cap F)$$

$$P(E/F) = \frac{P(E \cap F)}{P(F)}$$

$$P(E/F) = \frac{P(F/E)P(E)}{P(F)}$$

$$P(E \cap F) = P(E)P(F), \text{ for } E \text{ independent of } F$$

Random Variable

- Random variable: X - a real valued function defined on S

Cumulative Distribution Function (CDF)

CDF $F()$ defined on random variable X :

$$F(a) = P(X \leq a), \quad -\infty < a < \infty$$

has properties

- 1) $F(a)$ is a non-decreasing function of a
- 2) $\lim_{a \rightarrow \infty} F(a) = 1$
- 3) $\lim_{a \rightarrow -\infty} F(a) = 0$

Example

- Experiment: Roll a pair of dice
- All outcomes $S = \{(1,1), (1,2), \dots (1,6), (2,1), (2,2), \dots (5,6), (6,6)\}$

Event $E = \{\text{all first dice values} = 1\} = \{(1,1), (1,2), (1,3), (1,4), (1,5), (1,6)\}$

Event $F = \{\text{all first dice values} \leq 2\} = \{(1,1), (1,2), \dots, (1,6), (2,1), \dots (2,6)\}$

Example

$$P(E) = \frac{6}{36}$$

$$P(F) = \frac{12}{36}$$

$$P(E \cap F) = P(\{(1,1), (1,2), (1,3), (1,4), (1,5), (1,6)\}) = \frac{6}{36}$$

$$P(F/E) = 1$$

$$P(E \cup F) = P(E) + P(F) - P(E \cap F) = \frac{6}{36} + \frac{12}{36} - \frac{6}{36} = \frac{12}{36}$$

$$P(E/F) = \frac{P(E \cap F)}{P(F)} = \frac{\frac{6}{36}}{\frac{12}{36}} = \frac{1}{2}$$

$$P(E/F) = \frac{P(F/E)P(E)}{P(F)} = \frac{1 * \frac{6}{36}}{\frac{12}{36}} = \frac{1}{2}$$

$$P(E \cap F) = \frac{6}{36} \neq P(E)P(F) = \frac{6}{36} \frac{12}{36}, \text{ } E \text{ and } F \text{ are not independent events}$$

Example

- Random variable: X - a real valued function defined on S
- Let the function be defined as the sum of the two dice values.

Example

$$P[X = 2] = P\{(1,1)\} = \frac{1}{36}$$

$$P[X = 3] = P\{(1,2), (2,1)\} = \frac{2}{36}$$

$$P[X = 4] = P\{(1,3), (2,2), (3,1)\} = \frac{3}{36}$$

$\vdots$

$$P[X = 12] = P\{(6,6)\} = \frac{1}{36}$$

$$F(b) = P[X \leq 3] = P\{(1,1), (1,2), (2,1)\} = \frac{3}{12}$$

Types of Random Variables

Discrete: $p_X(a) = P(X = a)$

- Bernoulli, Binomial, Poisson

Continuous: $P[X \in A] = \int_A f_X(x)dx$

- Gaussian / Normal

Facts

- Every random variable has a distribution.
- In the discrete case, this distribution is called the probability mass function: $p_X(x)$
- In the continuous case, this distribution is called the probability density function: $f_X(x)$
- The distribution sums (discrete) or integrates (continuous) to 1 for the input domain x .

Bernoulli & Binomial Random Variables

- Bernoulli: Success / failure

$$p_X(0) = P[X = 0] = 1 - p$$

$$p_X(1) = P[X = 1] = p$$

- Binomial: n independent trials of Bernoulli random variable, (n,p)

$$p_X(k) = P(X = k) = \binom{n}{k} p^k (1 - p)^{n-k}, \quad k = 0, 1, 2, \dots, n$$

Poisson Random Variable

- Used to model image sensor shot noise

$$p_X(i) = P[X = i] = e^{-\lambda} \frac{\lambda^i}{i!}, \quad i = 0, 1, 2, \dots$$

Gaussian / Normal Random Variable

- ***Occurs naturally in nature.

$$f_X(x) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right), \quad x \in \mathbb{R}$$

Expectation

- Expectation of a random variable X : $E[X]$
- Expectation of a function of a random variable X : $E[g(X)]$

Expectation

- Discrete random variable X :

$$E[X] = \sum_x x p_X(x) = \sum_i x_i P(X = x_i)$$

- Continuous random variable X :

$$E[X] = \int_{-\infty}^{\infty} x f_X(x) dx$$

Expectation for the Function of a Random Variable X

- Discrete: $g(X)$

$$E[g(X)] = \sum_x g(x) p_X(x) = \sum_i g(x_i) P(X = x_i)$$

- Continuous: $g(X)$

$$E[g(X)] = \int_{-\infty}^{\infty} g(x) f_X(x) dx$$

Commonly Encountered Functions of a Random Variable

- Shift and Scale of random variable X : $g(X) = aX + b$
- Variance of random variable X : $g(X) = (X - \mu)^2$

Joint Random Variables

- Previously, the CDF for random variable X :

$$F(a) = P(X \leq a), \quad -\infty < a < \infty$$

- Now, the CDF for the jointly distributed random variables X and Y :

$$F(a, b) = P(X \leq a, Y \leq b), \quad -\infty < a, b < \infty$$

Joint Probability Mass / Density Function

- Joint probability mass function

$$p_{X,Y}(x, y) = P[X = x, Y = y]$$

- Joint probability distribution function

$$P[X \in A, Y \in B] = \int_B \int_A f_{X,Y}(x, y) dx dy$$

Joint Probability Mass / Density Function

- Marginal (discrete)
$$p_X(x) = \sum_y p_{X,Y}(x, y)$$
$$p_Y(y) = \sum_x p_{X,Y}(x, y)$$
- Marginal (continuous)
$$f_X(x) = \int_B f_{X,Y}(x, y) dy$$
$$f_Y(y) = \int_A f_{X,Y}(x, y) dx$$

Expectation of $g(X,Y)$

- Discrete: $g(X,Y)$

$$E[g(X, Y)] = \sum_y \sum_x g(x, y)p_{X,Y}(x, y)$$

- Continuous: $g(X,Y)$

$$E[g(X, Y)] = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} g(x, y)f_{X,Y}(x, y)dx dy$$

Commonly Encountered Functions of Random Variables

- $g(X,Y) = X + Y$
- $g(X,Y) = aX + bY$
- $E[g(X,Y)] = E[X+Y] = E[X] + E[Y]$
- $E[g(X,Y)] = E[aX+bY] = aE[X] + bE[Y]$

Independent Random Variables

- The random variables X and Y are independent if for ALL a, b :

$$P[X \leq a, Y \leq b] = P[X \leq a]P[Y \leq b]$$

- For X and Y discrete:

$$p_{X,Y}(x, y) = p_X(x)p_Y(y)$$

- For X and Y jointly continuous:

$$f_{X,Y} = f_X(x)f_Y(y)$$

Useful Results When X and Y are Independent

- Let X and Y be independent random variables.
- $E[g(X)h(Y)] = E[g(X)]E[h(Y)]$
- $\text{Var}(X+Y) = \text{Var}(X) + \text{Var}(Y)$
- $\text{Cov}(X, Y) = 0$

Covariance

- The covariance measures the relationship between random variables - i.e. how do the random variables “vary” with respect to one another.
- For two random variables X and Y , the covariance is defined as: $E[(X - \text{mean}X)(Y - \text{mean}Y)] = E[XY] - (E[X]E[Y])$
- The covariance can be positive, negative or 0.
- If X and Y are independent, the covariance is 0.
- If the covariance is 0, X and Y are not necessarily independent though.

Transforming Joint Random Variables

Let $X_1 = X$ and $X_2 = Y$, with distribution $f_{X_1, X_2}(x_1, x_2)$

Define $Y_1 = g_1(X_1, X_2)$ and $Y_2 = g_2(X_1, X_2)$

g_1 and g_2 have continuous partial derivatives.

What is the new distribution $f_{Y_1, Y_2}(y_1, y_2)$?

Transforming Joint Random Variables

$$f_{Y_1, Y_2}(y_1, y_2) = f_{X_1, X_2}(x_1, x_2) \left| \begin{array}{cc} \frac{\partial g_1}{\partial x_1} & \frac{\partial g_1}{\partial x_2} \\ \frac{\partial g_2}{\partial x_1} & \frac{\partial g_2}{\partial x_2} \end{array} \right|^{-1}$$

Then, solve for x_1 and x_2 in terms of y_1 and y_2 and substitute .

i . e . $x_1 = h_1(y_1, y_2)$ and $x_2 = h_2(y_1, y_2)$

Generalization

- We have shown results for 2 random variables, X and Y.
- However, everything (previously stated) for 2 random variables, can easily be generalized to n random variables: $\{X_1, X_2, \dots, X_n\}$

Limit Theorems

- There are two main theorems that result from what we have learned so far:
- 1) Law of Large Numbers
- 2) Central Limit Theorem

Law of Large Numbers

Suppose we sample from an IID (independent, identically distributed) distribution: $X_1, X_2, \dots, X_n$

Let:
$$X = \frac{X_1 + X_2 + X_3 + \dots + X_n}{n}$$

Then: $\lim_{n \rightarrow \infty} X = \mu$, where $E[X_i] = \mu$

Central Limit Theorem

Suppose we sample from an IID distribution with mean and variance

$$E[X_i] = \mu, \quad E[(X_i - \mu)^2] = \text{Var}(X_i) = \sigma^2$$

Then, ***

$$\lim_{n \rightarrow \infty} \frac{(X_1 + X_2 + \dots + X_n - n\mu)}{\sigma\sqrt{n}} \rightarrow N(0,1)$$

Stochastic Process / Random Process

- A stochastic process / random process is a time indexed sequence of random variables.

$[X(t), t \in T]$, where $X(t)$ is a collection of random variables.

- Stochastic processes are widely used in math, engineering, economics, physics ...

Conditional Probability and Associated Conditional Probability Mass and Conditional Probability Density Function

- Previously, conditional probability: $P(E/F) = \frac{P(EF)}{P(F)}$
- Conditional probability mass function: $p_{X/Y}(x/y) = \frac{p_{X,Y}(x,y)}{p_Y(y)}$
- Conditional probability density function: $f_{X/Y}(x/y) = \frac{f_{X,Y}(x,y)}{f_Y(y)}$

Conditional Expectation

$$E[X/Y = y] = \sum_x x p_{X/Y}(x/y)$$

$$E[X/Y = y] = \int_{-\infty}^{\infty} x f_{X/Y}(x/y) dx$$

The Power of Conditioning

- Conditioning, allows you to introduce additional information into a problem.
- In probability space, we conditioned on events, allowing us to compute the probability for event A, given that event B had occurred.
- With random variables, conditional expectation allows us to improve the estimate of the mean of the random variable, given knowledge of it's interaction with other random variables.
- We now show an example, where the interaction involves the total number of occurrences.

Example

- The poisson random variable counts the number of events that occur in a fixed interval of time.
- The probability mass function returns the probability of k events occurring, in a fixed interval of time.
- Suppose we observe that n total events have occurred for two independent poisson random variables X and Y . i.e. $X+Y = n$.
- Now, suppose we want to know the average number of events that occurred for the poisson random variable X , given the above information.
- How do we solve this?

Example

- Initial thought: Compute $E[X]$, since X and Y are independent.
- Rebuttal: It is true that the random variables are independent. With no additional knowledge, we would expect the frequency of occurrence for X would be it's mean, for it's given rate parameter.
- However, we know that n events have occurred between the two random variables.
- $\Rightarrow$ We need to make use of this information, to get a better estimate.

Example

- Problem formulation: $E[X | X + Y = n]$
- Since we want to compute the conditional expectation over X , we need the conditional probability mass function:

- $$P[X = k | X + Y = n] = \frac{P[X = k, X + Y = n]}{P[X + Y = n]}$$

$$P[X = k | X + Y = n] = \frac{P[X = k, Y = n - k]}{P[X + Y = n]}$$

Example

- Now, applying the knowledge that X and Y are independent:

$$P[X = k | X + Y = n] = \frac{P[X = k]P[Y = n - k]}{P[X + Y = n]}$$

- We can now substitute for all three terms, by applying the definition of the poisson random variables for X , Y and $Z=X+Y$, to obtain the conditional probability mass function.
- Finally, we apply the conditional expectation equation to the conditional probability mass function.

Conclusion

- These slides are meant to be a quick refresher for the fundamentals of probability.
- We have touched upon the highlights, focusing on the core results.
- Many ideas from deep learning, result from these simple and basic fundamentals.

Examples

- Unsupervised learning via Flow results from the idea of transforming distributions / random variables.
- Unsupervised learning via Diffusion is rooted in parameterized Markov chains (not discussed here).
- The latent vector z in supervised and unsupervised learning is based on a multivariate normal.
- Monte Carlo rollouts in reinforcement learning simulations are rooted in the law of large numbers.
- Image denoising using deep networks relies on a shot noise model rooted in the poisson distribution.

Examples

- Mutual information measures how much one random variable contains about another.
- Entropy measures the amount of uncertainty or randomness, in a random variable.
- KL Divergence measures the similarity / dis-similarity between two probability distributions.